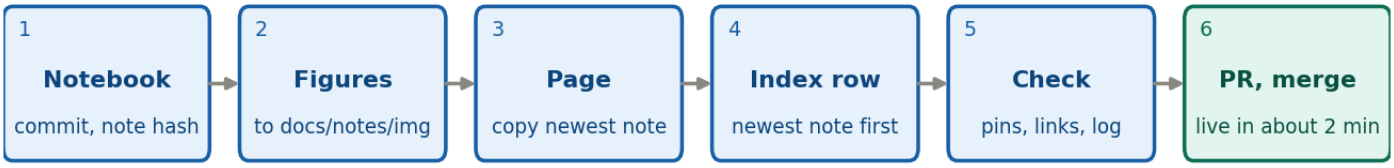
Creating a note

Developer quick-sheet · hand-written notes in `docs/notes/` · version 7, 21 October 2026

For a note you write yourself: an HTML page in `docs/notes/` and one row in the front-page table. (Student notes are Markdown in `notes/src/`; see the box at the bottom and the student cheat-sheet.) Start from `develop`, on a branch: `git switch -c feature/note-NN-short-name` (NN is the next note number, which student notes share: `python tools/build_notes.py next` prints it; a follow-up takes a letter, like 2a).



1. **Notebook.** Do the analysis in `data_analysis/R76/notebooks/`, run it top to bottom, and commit it with any library change. Note the short hash: `git rev-parse --short HEAD`. Re-running some notebooks leaves metadata churn, so read `git diff` before committing. Never run `07221203_2025_dump1_noise.ipynb` without approval (it rewrites `data_analysis/archives/good_noise.h5`).
2. **Figures.** Save the PNGs from the executed notebook into `docs/notes/img/`, named for the series and the content, for example `dump1_bstd_time.png`.
3. **Page.** Copy the newest note, `note-02a-bstd-vs-time.html`, to `docs/notes/note-NN-slug.html`. Edit the `<title>`, the `<h1>` (write it as a question), the meta line (`Note NN · Author(s) · date · Status`), and the `<nav class="outline">` with an `id` on every `<h2>`. Sections: Code Repository, Data and setup, The question, then the headings your story needs (one section per result with a `<figure>`, alt text and a caption; what it means and what could be wrong; next steps). Student guide 3 shows three notes with their headings.
4. **Pin the code.** Every link to a notebook or to `python/` uses the commit hash, never a branch name: under the repository URL, `blob/<hash>/<path>` for a file and `tree/<hash>/<folder>` for a folder. Check each one: `git cat-file -e <hash>:<path>` prints nothing when the file exists.
5. **Index row.** Add one `<tr>` to the Analysis Notes table in `docs/index.html`, **newest note first**. Columns: Note #, Link, Date, Author, Title, Description, Keywords, Status. The status badge is `<span class="status in-progress">` (or `complete`, `superseded`, `abandoned`, `brainstorm`).
6. **Changelog and checks.** Add a line under `[Unreleased]` in `CHANGELOG.md` (a new note is a patch-level change). Open the page from disk in a browser: figures show, links work. If you touched `python/`, run `cd python && python -m pytest -q`.
7. **PR.** `git push -u origin feature/note-NN-short-name`, then `gh pr create --base develop --fill`. When the checks are green, merge: the page is live in about two minutes.

Updating a note is the same loop, on a new branch from `develop`. Check its status first: a **Complete** note is left alone (a follow-up takes a letter, like 2a after 2), an **In progress** note can change freely, and a note that a later one replaces becomes **Superseded (reference/link)** with a link to the newer one. If you re-ran anything, re-pin the links to the new hash and replace the PNGs. Update the date and status in the page's meta line **and** in its row of `docs/index.html` (the row is hand-edited, so the two can drift). A wrong result stays in the note, corrected and dated; never delete a published note.

Rules. A **Complete** note gets only a minimal correction, and only when a library rename makes it wrong. A breaking library change means updating the notebooks that use it, or marking them deprecated (the red box described on the front page). Any whole-trace spectrum must skip the leading glitch samples. A *segment* is a run of same-trigger events; a *run* is a whole data-taking period.

Tip

Reviewing a student note. The PR into `develop` shows a green check and nothing is public. Change `id: "TBD"` to the next note number: the *Student notes* check posts it on the PR as a suggestion on the `id` line, so press **Commit suggestion** (no checkout; `python tools/build_notes.py next`

prints the same number, in the same numbering as the hand-written notes), and merge. With two note PRs open at once both are offered the same number: merge one, then re-run the check of the other to refresh its suggestion. The bot builds the page and its row in the Analysis Notes table, placed by number among the hand-written rows, and it is live in 2 to 3 minutes. When a student updates a published note, check the header still has its note number: `TBD` passes the PR check, but the build refuses it and nothing is published until you fix it. To remove a note, delete its source file in a PR; the next build removes the page.